CS-011 Inventory, Storage & Transport

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Title	Inventory, Storage & Transport
Revision	E
Status	Draft, pending Lead signature (storage-map locations to field-verify)
Effective date	2026-10-10
Supersedes	The stale Master Tracker Inventory sheet as a live record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	2026-10-10
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Added a reference to the app's Stock tab and noted its location field is freeform today, not a controlled vocabulary, so it doesn't yet enforce the storage map. Language pass.
C	2026-08-03	Claude	§7.3 storage map: locations become records with their own labels, so the location field is a controlled value rather than freeform text (CS-001 §7.10).
D	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). §7 dropped its "(outline)" label and its five run-on items became titled subsections; the storage-map paragraph was split so the rule, the mechanism and the payoff read separately. No rule changed.

Rev	Date	Author	Description of change
E	2026-10-10	Claude (doc overhaul, requested by Simon Starbuck)	§5 is now the app's restock table (Material type, unit, supplier, lead time) and says how a lead changes a minimum. §7.1 walks the shelves the way the app does and sends reorders to #purchasing-slackbot, since #purchasing was retired 2026-10-07. §7.3 uses the app's site names and Inventory → Boards. New §7.6 gives resin storage temperatures from the TDSs. §8 adds a no-chemicals-in-a-fridge check; §8 and §9 name the app records. Scope adds foam, core and donated stock. Tacky tape supplier set to Airtech AT-200Y (Simon, 2026-10-10). Crystallized 105 points to CS-008 rather than repeating it. §6 and the three disputed §5 minimums are unchanged, held for Simon. Register R-011-03, R-011-05..15, R-011-30 (§7.2 retitled), R-011-31 (Flammables cabinet row), R-000-09, R-008-10, R-014-05. The held spots (§5 minimums, §6) are marked.

1. Purpose

Our SN5 inventory sheet was written at season start and then never touched. MEKP sat flagged "RE-ORDER" for months with no owner and no trigger, and when the October Easy Composites order got stuck in customs we had no buffer and lost weeks of layups (PP-02). Storage was purely reactive: parts got evicted from the etch locker, molds blocked everyone's boxes at RFS (PP-10). This doc sets minimum stocks, gives them an owner, and maps where things live.

2. Scope

Consumables, resins and chemicals, cloth, tooling board, foam and core stock, molds, finished parts, and donated or sponsor material: where they live, when they get reordered, how they move. Receiving a delivery into the app and the lot records it makes are CS-014.

3. References

Ref	Document	Where
R1	PP-02, PP-10 RCAs	FEB-COMP-PP-001
R2	CS-012 (ordering process + lead times)	CS-012
R3	CS-001 (labels carry home locations)	CS-001
R4	SDS chemical storage sections	`03 Datasheets/`
R5	The app's Inventory tab (Storage map, Materials list, Boards)	`06 Composites App/app/README.md`
R6	CS-014 (receiving, lots, opening and emptying containers)	CS-014
R7	Resin TDS storage sections: IN2/AT30, XCR, West System 105 with 205/206/209	`03 Datasheets/EC-TDS-IN2-Infusion-Resin.pdf`, `03 Datasheets/EC-TDS-XCR-Epoxy-Coating-Resin.` `03 Datasheets/west-system_slow-epoxy-hardener` `03 Datasheets/105_205-207-Combined.pdf`, `03 Datasheets/105-209-Epoxy-Resin-1.pdf`

4. Definitions

- **Min-stock**: the level that triggers a reorder, sized so the order lands before stock-out at the supplier's lead time.
- **Stock walk**: the monthly check of every shelf against its record (§7.1). The app records it per shelf as Contents last confirmed and Confirmed by.
- **Home location**: the storage record (`BIN-SN6-###`) an object belongs on. Molds and material lots carry it in their Home location field.
- **Lot**: one recorded container or roll (`FAB-SN6-###`, `RSN-SN6-###`, `CON-SN6-###`); a consumable lot is one record with a count. CS-014 owns lots.
- **Material type**: the app's name for a kind of material across every container of it, for example `IN2`, `AT30` or `WEST-105`. Restock rules and a shelf page's grouping both key on it.
- **Restock rule**: one row of the §5 table as the app holds it: Material type, minimum, unit, supplier and lead time.
- **How full**: a lot's level, Full, Half, Low or Empty. An Empty container counts as zero toward the minimum; an opened one counts as one container (`app/inventory.js`, `restockOnHand`).

5. Minimum stock: what we never run out of

These are starting values, tuned with usage data. Sizing rule (§4): min stock = (burn rate × supplier lead time) + one job's worth. Worked examples for the three items that actually stop a layup; the rest of the table starts as one-job buffers, tuned with WO usage data:

- **IN2 + AT30:** about 1.95 kg mixed per large infusion (team estimate; no logged source); at the 2-infusions/week peak that is ~4 kg/week. Easy Composites lead 6 weeks (CS-012) → running dry mid-season costs ~6 weeks of infusions. One unopened kit (~4 kg, team estimate) 1 peak week: it is the alarm, not the buffer. When the walk finds only the unopened kit, the order is already late, so trigger at opened kit + 1 unopened, and place the big seasonal orders per CS-012 §7.4 so the walk never trips this.
- **XCR:** ~1 kit seals 2–3 mold sections (tune from WO records); molds queue 2/week in peak → 1 kit 1 week. Same logic: 1 unopened kit minimum, ordered with every EC basket.
- **Tacky tape:** every bagged job uses 1–2 rolls of perimeter; 2 jobs/week → ~3 rolls/week; domestic lead ~1 week → 6 rolls 2 weeks of peak burn.

Material (Material type)	Minimum	Unit	Supplier	Lead time	Why it blocks
IN2 infusion resin (`IN2`)	2 on hand, one may be open	kit	Easy Composites	42 days	Every infusion; 6-week EC lead
AT30 slow hardener (`AT30`)	2 on hand, one may be open	kit	Easy Composites	42 days	Pairs with IN2
West 105 + 206 (`WEST-105`, `WEST-206`)	½ tank + 1 can [held for Simon: R-011-04; the app holds 1 tank]	tank, can	West System	14 days	Every wet layup
XCR (`XCR`)	1 [held for Simon: R-011-04; the app holds 2]	kit	Easy Composites	42 days	Every mold
VB160 bag film (`VB160`)	1	roll	Easy Composites	42 days	Everything bagged
Peel ply (`PEEL-PLY`)	1	roll	Easy Composites	42 days	Everything bagged
Flow mesh (`FLOW-MESH`)	1	roll	Easy Composites	42 days	Everything infused
Tacky tape (`TACKY-TAPE`)	6	roll	Airtech (AT-200Y, `03 Datasheets/AIRTECH-TDS-AT-200Y-Sealant-Tape.pdf`)	7 days (team planning value)	Everything bagged; team
Nitrile gloves (`GLOVES-NITRILE`)	2	box per size	McMaster	7 days	Everything
Mixing cups (`MIXING-CUPS`)	50	cup	McMaster	7 days	Everything
Mixing sticks (`MIXING-STICKS`)	50	stick	McMaster	7 days	Everything
Respirator cartridges, A + P100 (`CARTRIDGE-A-P100`)	2	set	McMaster	7 days	Safety-blocking
195 twill cloth (`195-TWILL`)	10 [held for Simon: R-011-04; the app holds 1 roll]	yd	Sigmatex (sponsor)	30 days	Most stacks

Lead times are the team's planning values from the app's restock table (`app/inventory.js`, `RESTOCK_SEED`);

Easy Composites' 42 days is the 6 weeks of §7.2 (PP-02).

The app holds this table as its restock rules, and those are the live values: the Running out card on Inventory → Storage map lists every Material type at or below its minimum. Until R-011-04 is decided, three rows differ from the app: it holds XCR at 2 kits, West 105 at 1 tank and 195 twill at 1 roll, and the Running out card uses the app's values. A lead tunes a minimum by editing `config/restock` in the Firebase console; there is no editor in the app. A change there is a change to this table, and the lead who makes it shall open a revision of this standard within two weeks (CS-000 §2, §7.2).

A Material type appears on Running out only once at least one lot carries it. The Receiving desk fills it from a typed name (CS-014 §7.3 step 6); rows filled by Arrived or Take all do not get one. For a lot that still has none, press Link materials on the Materials list (CS-014 §7.7 step 3) or set it with Edit on the lot.

6. Safety

Chemical storage per each SDS (R4): resins/hardeners in the chemical boxes/barrel, **never in food fridges** (2026-02-17 incident); flammables (Airtac, acetone, clear coat) in the flammables cabinet; segregate hardeners from resins on different shelves.

[held for Simon: co-storage and the flammables list, REGISTER R-011-01, R-011-02]

7. Procedure

7.1 Monthly stock walk

The walk takes about 15 minutes (team estimate). An IA or first-year is the right owner; it is a good first responsibility.

1. The lead names the stock-walk owner at the start of the season.
2. Once a month, open Inventory → Storage map and walk each shelf: open its card, check the list against what is on the shelf, and press Confirm contents (Confirm on the map card does the same).
3. Fix what is wrong as you go: Move for something on the wrong shelf, How full on a container, and the shelf's Add here card for anything missing.
4. Open the Running out card on the storage map. If any row lacks "on order", press Add N to a purchase, pick A new purchase or an open purchase under Add to, and press Add to purchase.
5. Post the purchase request in #purchasing-slackbot the same day (CS-012 §7.1).

Between walks, open one container of a material at a time and mark an emptied container Empty rather than deleting it: the cure buy-off pre-fills the most recently opened lot, and the reorder count needs to see the empty one (CS-014 §7.8).

7.2 Order timing: when to order resin and cloth

One big fall order plus one spring top-up beats four rushed baskets.

1. Place Easy Composites orders at least 6 weeks before the material is needed: UK shipping plus customs (PP-02).
2. Order on the calendar in CS-012 §7.4: one big fall order and one spring top-up.

7.3 Where things are stored (storage map)

Where to store or keep molds, parts, cloth, chemicals and consumables:

Object	Site (as the app names it)	Rule
Molds	RFS container	Never in front of the knaack or screw boxes (PP-10)
Foam and tooling-board stock	RFS container	Board stock goes on Inventory → Boards (step 1)

Object	Site (as the app names it)	Rule
Finished parts	Jacobs basement	Never the etch locker (2026-04-20 agreement); keycard access
Cloth	Dry sealed bin	None beyond the site
Resins and hardeners	No app Site names them yet; per §6	Never in a food fridge (§6)
Flammables (what goes in it: §6)	Flammables cabinet	Rated cabinet only
Consumables	General Box	At RFS
Anything else	Other	Name the location after the place

These sites still need a field check against the real shelves.

1. Log tooling boards and offcuts under Inventory → Boards (R5); that is where real board inventory lives day to day.
2. Every shelf, bin and rack gets a storage record (+ Location on the storage map, with its Site set from the table above) and a printed label on its front edge (CS-001 §7.10), and every mold, part and material lot points at one.
3. To record a move, scan the object, tap Move, scan the shelf. To put several things on one shelf, open the shelf and use Move here (scan).

Since 2026-08-03 this is enforceable: "where is the seat mold" is a lookup rather than a Slack ask, and the monthly walk checks reality against a controlled value instead of whatever string somebody typed.

A shelf's page folds identical containers into one row by Material type, showing the count (×N), their states, the first expiry and the price. Open the row to see each container: it lists the FEB ID and the EH&S edge code, the last 12 characters printed down the right edge of the UC EH&S tag, so the screen matches the jug in your hand. Print the shelf's page for the walk sheet; every row prints open.

7.4 Transport

1. Molds over ~1 m²: two people, and a vehicle that actually fits them. Plan it, don't discover it.
2. Parts travel padded and labeled.

The SN5 budget shows a U-Haul was needed for mold moves.

7.5 Season handoff

1. At season end, the stock-walk owner and the lead produce the "what we actually have" list for the incoming lead (the 6/15 handoff gap): on Inventory, press Export, then Copy for Sheets on Everything on every shelf.
2. Paste it into a new sheet in the season's Drive folder.

7.6 Storage temperature and shelf life (resins, hardeners and XCR)

Keep every resin and hardener container closed. The figures below are datasheet figures (R7), not team holds. Where each container lives is §6 and §7.3.

Material	Store at	Shelf life	Source
IN2 resin and AT30 hardener	10–25 °C (50–77 °F), tightly sealed	12 months when stored correctly	`03 Datasheets/EC-TDS-IN2-Infusion-Resi`
XCR resin and hardener	20–25 °C (68–77 °F), tightly sealed	12 months when stored correctly	`03 Datasheets/EC-TDS-XCR-Epoxy-Coating`

Material	Store at	Shelf life	Source
West System 105 with 205, 206 or 209	Room temperature, containers closed	Usable for many years with proper storage	03 Datasheets/west-system_slow-epoxy-h 03 Datasheets/105_205-207-Combined.pdf 03 Datasheets/105-209-Epoxy-Resin-1.pd

1. Do not let West 105 freeze and thaw repeatedly: it crystallizes (03 Datasheets/west-system_slow-epoxy-hardener_2 Crystallized 105 is warmed to 125 °F and stirred before use; CS-008 §7 step 8 owns that step.
2. After long storage, check the West pumps' metering and mix a small test batch to confirm it cures (same TDS).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Stock walk done	Monthly during build season	Confirm contents on every active shelf; the storage map's "need a walk" chip reads 0 (it counts shelves not confirmed in 30 days, `app/inventory.js`)	Each shelf's Contents last confirmed and Confirmed by
No layup blocked by stock-out	0 per season	WO delay notes	WO timeline
Molds in home locations	100% at walk	Walk	Each mold's Home location; Export → Locations
No chemical lot on a fridge-type location	0 at walk	Open every location whose Type is fridge and check it holds no resin, hardener or other chemical lot; the app does not warn	Walk

9. Records

Each shelf's Contents last confirmed and Confirmed by fields; each mold's Home location; Export → Locations (Copy for Sheets or Download .csv) as the walk summary; Restock purchases in the Budget tab and their #purchasing-slackbot requests.



Figure 1: Where things are stored, by site